Constrained Graph Kinematics: Biomechanical Range-of-Motion for Occluded Hand Gesture Telemetry

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ABSTRACT

Three-dimensional (3D) hand gesture telemetry in surgical navigation and industrial teleoperation degrades when handheld instruments obscure phalangeal joints or when medical-grade latex and nitrile gloves introduce specular reflections and elastic stiffness. Standard on-device hand trackers such as MediaPipe Hands and Apple ARKit regress unconstrained 3D Cartesian coordinates in $\mathbb{R}^{21 \times 3}$. When 30% to 70% of joints lose visual line-of-sight, direct coordinate regressors produce anatomically impossible poses characterized by phalangeal bone stretching (15–27 mm error), finger collapse toward the palm, and dorsal joint hyperextension. Conversely, high-capacity occlusion-aware mesh transformers require tens of millions of parameters (>100 MB), exceeding the compute and memory budgets of embedded edge processors. We present **Constrained Graph Kinematics**, a two-stage spatio-temporal graph distillation and differentiable forward-kinematic framework. First, we design a 520,189-parameter **Spatio-Temporal Graph Autoencoder (`SpatioTemporalGAE`) Teacher** that combines visibility-gated 3-layer, 4-head Graph Attention over the 21-joint hand topology with a Bidirectional Temporal Gated Recurrent Unit (GRU) across T=8 sliding windows. Second, we compress the Teacher via a multi-objective rotational and positional **Knowledge Distillation** loss into a 78,398-parameter **Edge Student (`StudentKinematicMLP`)** (6.64× parameter reduction, 137.3× FLOPs reduction to 163.4 KFLOPs/frame). Both Teacher and Student map latent features into continuous 6D rotation representations on $\mathrm{SO}(3)$, clamp Euler flexion, twist, and abduction angles via a smooth hyperbolic tangent (`BiomechanicalRoMClamper`) layer, and reconstruct 3D joint positions through an exact **Differentiable Forward Kinematics (`DifferentiableFK`)** chain. Across 30%, 50%, and 70% surgical tool occlusion on our held-out test split (960 sequences, 7,680 frames) and our custom gloved benchmark (120 sequences), the Teacher achieves 3.33 mm, 3.79 mm, and 6.35 mm Mean Per-Joint Position Error (MPJPE), while the distilled Student achieves 4.08 mm, 5.05 mm, and 8.36 mm MPJPE—reducing error by 66.4\% at 50% occlusion compared to an unconstrained coordinate-regression MLP baseline (15.03 mm) while guaranteeing 0.0000 mm **bone-length error** and 100.0% **biomechanical validity** on every frame. Compiled to a static 433.2 KB ONNX Opset 17 graph, the Student executes in 0.121 ms/frame (8,256 FPS) on x86_64 CPU, 0.314 ms (3,185 FPS) on Apple Silicon MacBook Air, 2.046 ms (488.8 FPS) on NVIDIA Jetson Nano TensorRT FP16, and 1.044 ms end-to-end on live 60 Hz iPhone ARKit UDP telemetry streams.

Index Terms— 3D Hand Pose Estimation, Differentiable Forward Kinematics, Spatio-Temporal Graph Attention, Continuous 6D Rotations, Biomechanical Constraints, Knowledge Distillation, Edge AI Telemetry.

I. INTRODUCTION

Estimating three-dimensional (3D) hand kinematics from monocular cameras, depth sensors, or sparse keypoint streams is an ill-posed inverse problem. Anatomically, the human hand skeleton forms a directed tree $G = (V, E)$ comprising $|V| = 21$ joints connected by $|E_{\text{tree}}| = 20$ rigid bone segments across five digits (thumb carpometacarpal [CMC], metacarpophalangeal [MCP], and interphalangeal [IP] joints; index, middle, ring, and pinky MCP, proximal interphalangeal [PIP], and distal interphalangeal [DIP] joints) anchored at the wrist root ($j = 0$). In clinical operating rooms, robotic teleoperation cells, and cleanroom assembly lines, optical hand telemetry encounters two compounding physical failure sources:

1. **Contiguous Phalangeal Occlusion by Handheld Instruments:** Grasping surgical forceps, scalpels, endoscopic probes, or power drivers blocks camera line-of-sight across contiguous distal subtrees (MCP → PIP → DIP → TIP), masking 30% to 70%

of the hand skeleton simultaneously.

2. **Latex and Nitrile Glove Domain Shift:** Medical and industrial gloves eliminate dorsal skin texture, fingernail boundaries, and chromatic cues while introducing two physical artifacts: (a) elastic material resistance that dampens finger flexion toward a semi-open equilibrium posture, and (b) directional specular reflections under overhead luminaires that induce temporally correlated tracking jitter and distal fingertip spikes up to 12.0 mm.

When visual features disappear under tool occlusion or specular glare, unconstrained coordinate regressors—networks whose output layer predicts 63 independent scalars $\hat{P} \in \mathbb{R}^{21 \times 3}$ trained via Euclidean Mean Squared Error (MSE)—fail in three ways. First, missing distal phalanges drift toward the palm centroid or origin, shrinking or stretching bone lengths by 4–27 mm. Second, independent per-joint coordinate predictions violate physiological Range-of-Motion (RoM) bounds, producing negative PIP/DIP hyperextension angles or unnatural lateral finger splay. Third, frame-independent inference amplifies high-frequency optical glare into severe second-order acceleration jitter.

Prior work has attempted to mitigate these failures either by training large vision transformers on hand-object datasets [1]–[4], [8], [9] or by adding soft biomechanical penalty losses during training [11]. However, transformer architectures such as HandOccNet [8] (37.2M parameters, 142 MB) and METRO [9] (102M parameters) require desktop GPUs and cannot execute at high frame rates on embedded edge processors such as the NVIDIA Jetson Nano or Raspberry Pi. Meanwhile, soft training-time penalty losses (Spurr et al. [11]) do not constrain the output layer at test time; under severe out-of-distribution occlusion, data likelihood gradients override the penalty priors, yielding stretched bones and invalid joint angles during inference.

Contributions

To resolve the trade-off between anatomical validity under severe occlusion and sub-millisecond edge execution, this report introduces four technical contributions:

1. **3D Graph-Space Tool Occlusion & Zero-Shot Glove Domain Adaptation (`data/`):** Bhavya formulated `SyntheticOcclusionTransform`, which combines 3D cylindrical instrument clipping ($R = 28.0$ mm) with contiguous distal-to-proximal kinematic subtree priority at exact 30%, 50%, and 70% occlusion rates, alongside `SyntheticGloveTransform`, which models 12\% elastic stiffness and temporally correlated AR(1) specular sensor jitter ($\rho = 0.65$, 4.5 mm baseline, 12.0 mm fingertip spikes) across 4,800 MANO sequences (38,400 frames) and a 120-sequence custom gloved benchmark.
2. **Hard-Constrained Differentiable & Bare-Metal Edge Forward Kinematics (`models/kinematics.py`):** Vrushank designed an ONNX-traceable kinematic layer combining Gram-Schmidt continuous 6D-to-$SO(3)$ orthogonalization [12], a C^1-smooth hyperbolic tangent angular clamper (`BiomechanicalRoMClamper`), and recursive hierarchical Forward Kinematics (`DifferentiableFK` in PyTorch and `EdgeFK` in pure NumPy). By mathematical construction on $SO(3)$, our architecture guarantees 0.0000 mm **bone-length error** and 100.0% **biomechanical validity** on every output pose.
3. **Mask-Aware Spatio-Temporal Graph Autoencoder Teacher (`models/teacher_st_gae.py`):** Abhishekh architected `SpatioTemporalGAE` (520,189 parameters), which imputes occluded nodes with canonical rest priors, propagates spatial context across a 21-node anatomical graph via three 4-head Graph Attention layers equipped with a learnable visibility key gate γ_h, and integrates temporal dynamics across $T=8$ frames via a Bidirectional GRU—reducing 50%-occlusion MPJPE by 38.9\% relative to our `SpatialOnlyGAE` ablation (3.79 mm vs. 6.20 mm).
4. **Multi-Objective Kinematic Distillation & Multi-Hardware Edge Deployment (`models/student_models.py`, `distillation/`, `deployment/`):** Kapish distilled the frozen Teacher into a 78,398-parameter `StudentKinematicMLP` (6.64× parameter compression, 137.3× FLOPs reduction), exported it to a 433.21 KB ONNX Opset 17 model with 0.000254 mm numerical parity, and benchmarked real-time execution across x86_64 PC (0.121 ms/frame), Apple Silicon MacBook Air (0.314 ms), NVIDIA Jetson Nano TensorRT FP16 (2.046 ms), Raspberry Pi 5 (3.052 ms), and 60 Hz iPhone ARKit UDP telemetry (1.044 ms end-to-end).

II. RELATED WORK

A. 3D Hand Pose & Occlusion Benchmarks

Modern 3D hand pose estimation builds upon calibrated multi-camera benchmarks fitted with the parametric **MANO** (*hand Model with Articulated and Non-rigid defOrmations*) mesh [13]. **FreiHAND** (Zimmermann et al. [1]) established cross-domain single-RGB evaluation across 134,200 frames using Mean Per-Joint Position Error (MPJPE) and Procrustes-Aligned MPJPE

(PA-MPJPE). To evaluate tracking under heavy object manipulation, **HO-3D** (Hampali et al. [2]) and **DexYCB** (Chao et al. [3]) captured RGB-D sequences of subjects grasping rigid YCB objects where up to 60% of the hand is obscured by held tools. **InterHand2.6M** (Moon et al. [4]) demonstrated that when identically textured phalanges overlap during bimanual interaction, local visual features become ambiguous, requiring structural kinematic priors to disambiguate joint assignments.

B. Graph Convolutional and Occlusion-Aware Architectures

Because the hand skeleton forms a natural graph, **HandGCN** (Ge et al. [5]) applied Chebyshev spectral graph convolutions ($g_\theta * x = \sum_{k=0}^{K-1} \theta_k T_k(\tilde{L})x$) to reconstruct 3D hand meshes, while **HOPE-Net** (Doosti et al. [6]) cascaded two Adaptive Graph Convolutional Networks with learnable adjacency matrices $\tilde{A}_{ada}$ to lift 2D keypoints to 3D. **Pose2Mesh** (Choi et al. [7]) decoupled 2D keypoint detection from 3D graph lifting to improve cross-domain generalization. However, HandGCN, HOPE-Net, and Pose2Mesh regress unconstrained 3D vertex/joint coordinates in $\mathbb{R}^{V \times 3}$, suffering 3.4–4.8 mm bone-length distortion when input keypoints are corrupted by occlusion. To recover occluded features, **HandOccNet** (Park et al. [8]) introduced Feature-Injecting and Self-Enhancing Transformers, and **METRO** (Lin et al. [9]) applied non-local self-attention across all 799 joint and mesh tokens, while **MeTRAbs** (Sárándi et al. [10]) predicted metric-scale 3D volumetric heatmaps. While accurate (9.1 mm PA-MPJPE on HO-3D for HandOccNet), these architectures require 37M–102M parameters and >30 ms/frame on desktop GPUs.

C. Biomechanical Constraints & Continuous Rotation Manifolds

While MANO [13] parameterizes hand meshes via axis-angle rotations $\theta \in \mathbb{R}^{16 \times 3}$, unconstrained axis-angle regression permits physiologically invalid joint hyperextension. **Spurr et al.** [11] proposed weakly supervised soft penalty losses on bone lengths (L_{bl}), palmar arch curvature (L_{palm}), and joint flexion/abduction hull boundaries ($L_{ja} = \sum_i \max(0, \underline{\theta}_i - \theta_i) + \max(0, \theta_i - \overline{\theta}_i)$). Because these penalties act only as additive training regularizers on an unconstrained $\mathbb{R}^{21 \times 3}$ output layer, test-time occlusion still induces bone stretching (1.9 mm) and invalid joint angles (8.6\% violation rate). When parameterizing rotations for forward kinematics, **Zhou et al.** [12] proved topologically that all $\leq$ 4D representations of $SO(3)$ (including 3D Euler angles and 4D unit quaternions) contain discontinuities that impede gradient descent, introducing a continuous 6D Gram-Schmidt representation ($\mathbb{R}^6 \to SO(3)$). Finally, lightweight edge trackers such as **MediaPipe Hands** (Zhang et al. [14]) execute at >200 FPS on mobile CPUs/NPUs (1.5M params) but rely on unconstrained coordinate regression, exhibiting 6.2 mm bone-length error and 35.2\% anatomical invalidity under tool occlusion.

Method / Architecture	Primary Venue	Representation & Constraint Type	MPJPE (mm) ↓	PA-MPJPE (mm) ↓	Bone-Length Error (mm) ↓	Biomechanical Validity (%) ↑	Params / Model Size (MB) ↓	Inference Latency / Throughput ↑
HandGCN [5]	CVPR 2019	Spectral GCN (3D Vertices, Soft Losses)	13.9–18.2	9.8–11.2	3.40	78.5\%	18.50M / 74.0 MB	28.0 ms / 35 FPS (GPU)
HOPE-Net [6]	CVPR 2020	Adaptive-GCN (Unconstrained 3D Coords)	26.8 (HO-3D)	11.0 (HO-3D)	4.80	71.0\%	25.20M / 96.0 MB	16.5 ms / 60 FPS (GPU)
Spurr et al. [11]	ECCV 2020	CNN + 3D Coords (Soft Penalty Losses)	22.4 (HO-3D)	9.9 (HO-3D)	1.90	91.4\%	23.60M / 90.0 MB	15.0 ms / 66 FPS (GPU)
HandOccNet [8]	CVPR 2022	Dual Transformers (FIT/SET) + MANO	24.0 (HO-3D)	9.1 (HO-3D)	0.80	94.2\%	37.20M / 142.0 MB	33.5 ms / 30 FPS (GPU)
MediaPipe Hands [14]	CVPRW 2020	Mobile CNN (Unconstrained 2.5D/3D)	19.5–28.4*	11.4–16.2*	6.20	64.8\%	1.50M / 5.50 MB	4.50 ms / 220 FPS (CPU)
`UnconstrainedStudentMLP`	Ours (Ablation)	Direct $\mathbb{R}^{21 \times 3}$ Regression (MSE, No FK)	15.03	12.34	5.8505	8.70\%	0.0785M / 0.30 MB	0.021 ms / 47,114 FPS (ORT)

Method / Architecture	Primary Venue	Representation & Constraint Type	MPJPE (mm) ↓	PA-MPJPE (mm) ↓	Bone-Length Error (mm) ↓	Biomechanical Validity (%) ↑	Params / Model Size (MB) ↓	Inference Latency / Throughput ↑
`SpatialOnlyGAE`	Ours (Ablation)	3-Layer GAT + 6D SO(3) + Hard FK (No GRU)	6.20	4.58	0.0000	100.0\%	0.4617M / 1.77 MB	0.590 ms / 1,696 FPS (CPU)
`Teacher ST-GAE`	**Ours (Teacher)**	**3-Layer GAT + Bi-GRU + Hard** tanh **FK**	3.79	2.11	0.0000	100.0\%	0.5202M / 1.99 MB	0.676 ms / 1,479 FPS (CPU)
`StudentKinematicMLP`	**Ours (Student)**	**Distilled MLP + 6D** SO(3) + **Hard FK**	5.05	3.43	0.0000	100.0\%	0.0784M / 0.30 MB	0.121 ms / 8,256 FPS **(ORT)**

\Table I: State-of-the-art comparison. Published literature metrics reflect HO-3D/FreiHAND benchmarks; our ablation, Teacher, and Student rows report exact held-out test metrics under 50% surgical tool occlusion and latex glove degradation (`results/metrics_summary.json`).*

III. MATHEMATICAL FORMULATION OF 3D GRAPH TRANSFORMS

To train and evaluate our models across controlled clinical and industrial failure modes without data leakage, Bhavya implemented a deterministic 3D MANO trajectory synthesis and domain-adaptation pipeline (`data/dataset_ingestion.py`, `data/transforms.py`, `data/dataloader.py`).

A. Canonical 21-Joint Hand Graph & Stratified Dataset Split

Let $G = (V, E)$ denote the 21-joint hand graph with vertex set $V = \{0, 1, ..., 20\}$, where node 0 is the wrist root and nodes 1–20 form five directed kinematic chains: Thumb ($0 \to 1 \to 2 \to 3 \to 4$), Index ($0 \to 5 \to 6 \to 7 \to 8$), Middle ($0 \to 9 \to 10 \to 11 \to 12$), Ring ($0 \to 13 \to 14 \to 15 \to 16$), and Pinky ($0 \to 17 \to 18 \to 19 \to 20$), with parent mapping $p(j) = \text{PARENT}_\text{INDICES}[j]$. The graph edge set E comprises the 20 parent-child kinematic bone edges plus 4 inter-MCP palmar arch edges $E_{palm} = \{(1, 5), (5, 9), (9, 13), (13, 17)\}$ and 21 self-loops. Each bone $j \in \{1, ..., 20\}$ possesses a fixed canonical length $L_j \in \mathbb{R}^+$ in millimeters (`CANONICAL_BONE_LENGTHS_MM`) and unit rest direction $\hat{u}_j^{rest} \in S^2$.

We generate $N = 4{,}800$ smooth multi-frame 3D trajectories of length $T = 8$ (38,400 total frames) across four random seeds (`seed42`, `seed72`, `seed92`, `seed112` in `data/processed/`) spanning five functional articulation archetypes (*Pinch, Power Grasp, Tripod Precision, Index Pointing, Open Flexion*). The corpus is partitioned via a strict **80% Train (**$3{,}840$ **sequences,** $30{,}720$ **frames) / 20% Held-Out Test (**960 **sequences,** $7{,}680$ **frames)** split (`data/splits/split_indices.json`). Additionally, `data/record_custom_gloved.py` generates an independent 120-sequence (960-frame) multi-tier gloved benchmark (`data/custom_gloved_dataset.npz`).

B. Zero-Shot Glove Domain Adaptation (`SyntheticGloveTransform`)

Protective latex and nitrile gloves alter raw 3D hand telemetry through two physical mechanisms:

1. **Kinematic Range Reduction (Elastic Stiffness):** The elastic glove membrane resists full finger flexion, pulling each bone vector toward the semi-flexed neutral glove equilibrium $\hat{u}_j^{rest}$ with damping coefficient $\alpha_{stiff} = 0.12$. Given clean ground-truth positions $P_{t,j}^{GT} \in \mathbb{R}^3$ and current unit bone direction $\hat{d}_{t,j} = (P_{t,j}^{GT} - P_{t,p(j)}^{GT}) / L_j$, the stiffened pose $\tilde{P}_{t,j}$ is constructed recursively from the wrist root $\tilde{P}_{t,0} = P_{t,0}^{GT}$:

$$\tilde{d}_{t,j} = \frac{(1 - \alpha_{stiff})\hat{d}_{t,j} + \alpha_{stiff}\hat{u}_j^{rest}}{\|(1 - \alpha_{stiff})\hat{d}_{t,j} + \alpha_{stiff}\hat{u}_j^{rest}\|_2}, \qquad \tilde{P}_{t,j} = \tilde{P}_{t,p(j)} + L_j \tilde{d}_{t,j} \tag{1}$$

1. **Temporally Correlated AR(1) Specular Sensor Jitter:** Optical and LiDAR trackers suffer correlated frame-to-frame depth drift on shiny latex surfaces plus localized specular glare spikes at the distal fingertips $V_{tip} = \{4, 8, 12, 16, 20\}$. We model the baseline noise via a first-order autoregressive process AR(1) with temporal correlation $\rho = 0.65$ and per-joint standard

deviation $\sigma_j = 4.5$ mm (5.625 mm on DIP joints $j \in \{3, 7, 11, 15, 19\}$, 0 mm on the wrist anchor $j=0$), superimposed with Bernoulli specular spikes ($p_{spike} = 0.20$, $\sigma_{spike} = 12.0$ mm):

$$\varepsilon_{t,j} = \rho\, \varepsilon_{t-1,j} + \sqrt{1 - \rho^2}\, z_{t,j}, \quad z_{t,j} \sim N(0, I_3) \tag{2}$$

$$P_{t,j}^{gloved} = \tilde{P}_{t,j} + \sigma_j\, \varepsilon_{t,j} + \mathbb{I}(j \in V_{tip})\, b_{t,j}\, \xi_{t,j}, \quad b_{t,j} \sim \text{Bernoulli}(p_{spike}), \quad \xi_{t,j} \sim N(0, \sigma_{spike}^2\, I_3) \tag{3}$$

C. 3D Volumetric Tool Clipping & Kinematic Subtree Masking (`SyntheticOcclusionTransform`)

To simulate a surgical scalpel, forceps, or industrial handle crossing the hand, `SyntheticOcclusionTransform` spawns a moving 3D cylindrical volume of radius $R_{cyl} = 28.0$ mm and unit axis $\hat{u}_{tool} \in S^2$ with time-varying center $c_t = (1 - \frac{t}{T-1})c_{start} + \frac{t}{T-1}c_{end}$ anchored near the distal phalangeal centroid. For each joint j at frame t with relative offset $r_{t,j} = P_{t,j}^{gloved} - c_t$, we compute the signed cylinder proximity and camera-ray (+Z) depth occlusion score $d_{t,j}^{vol}$:

$$d_\perp(t,j) = \|r_{t,j} - (r_{t,j} \cdot \hat{u}_{tool})\hat{u}_{tool}\|_2, \quad d_\parallel(t,j) = |r_{t,j} \cdot \hat{u}_{tool}|, \quad d_z(t,j) = \max(0, P_{t,j,z}^{gloved} - c_{t,z}) \tag{4}$$

$$d_{t,j}^{vol} = d_\perp(t,j) + 0.25\max(0, d_\parallel(t,j) - 45.0) - 0.20\, d_z(t,j) - R_{cyl} \tag{5}$$

To ensure that occlusion propagates along physically contiguous finger subtrees from distal tip to proximal knuckle (TIP → DIP → PIP → MCP), each finger $f \in \{1, ..., 5\}$ is assigned a mean distal proximity $\bar{d}_{t,f} = \frac{1}{3}\sum_{r=0}^{2} d_{t,chain_f[r]}^{vol}$, and each joint $j = chain_f[r]$ ($r \in \{0, 1, 2, 3\}$ ordered distally) receives priority score:

$$s_{t,j} = \bar{d}_{t,f} + 0.35\, d_{t,j}^{vol} + 6.0\, r, \quad s_{t,0} = +10^9 \tag{6}$$

Given target occlusion rate $\eta \in \{0.30, 0.50, 0.70\}$, the exact $k_{mask} = \text{round}(\eta \cdot 21)$ joints with the lowest scores $s_{t,j}$ are masked ($m_{t,j} = 0$, $P_{t,j}^{obs} = 0$), while the remaining $21 - k_{mask}$ visible joints retain $m_{t,j} = 1$ and $P_{t,j}^{obs} = P_{t,j}^{gloved}$.

IV. 6D CONTINUOUS ROTATIONS & DIFFERENTIABLE FORWARD KINEMATICS

To guarantee that predicted 3D skeletons never exhibit bone-length distortion or physiological hyperextension, Vrushank implemented a differentiable kinematic manifold in `models/kinematics.py`.

A. Singularity-Free 6D-to-$SO(3)$ Orthogonalization (`rot6d_to_matrix`)

Following Zhou et al. [12], each joint $j \in \{0, ..., 20\}$ is parameterized by a continuous 6D vector $d_j = [a_{j,1}^\top, a_{j,2}^\top]^\top \in \mathbb{R}^6$, where $a_{j,1}, a_{j,2} \in \mathbb{R}^3$. The mapping $g_{GS}: \mathbb{R}^6 \to SO(3)$ constructs an orthonormal right-handed rotation matrix $R_j \in SO(3)$ via Gram-Schmidt orthogonalization:

$$b_{j,1} = \frac{a_{j,1}}{\|a_{j,1}\|_2}, \quad u_{j,2} = a_{j,2} - (b_{j,1} \cdot a_{j,2})b_{j,1}, \quad b_{j,2} = \frac{u_{j,2}}{\|u_{j,2}\|_2}, \quad b_{j,3} = b_{j,1} \times b_{j,2} \tag{7}$$

$$R_j = \begin{vmatrix} | & | & | \\ b_{j,1} & b_{j,2} & b_{j,3} \\ | & | & | \end{vmatrix} \in SO(3), \quad R_j^\top R_j = I_3, \quad \det(R_j) = +1 \tag{8}$$

In `models/kinematics.py`, degenerate near-zero or collinear vectors ($\|a_{j,1}\|_2 < 10^{-6}$ or $\|u_{j,2}\|_2 < 10^{-6}$) are guarded via `torch.where` fallbacks onto canonical basis axes e_1, e_2, e_3, ensuring that $R_j \in SO(3)$ holds for every possible tensor input using pure ONNX-standard operators.

B. C^1-Smooth Euler Extraction & tanh Biomechanical RoM Clamping (`BiomechanicalRoMClamper`)

While Gram-Schmidt orthogonalization constrains each joint to $SO(3)$, unconstrained rotations on $SO(3)$ still allow a finger joint to bend backward by 90^\circ (dorsal hyperextension) or twist axially. Let $R_j = R_z(\theta_{j,z})R_y(\theta_{j,y})R_x(\theta_{j,x})$ denote the intrinsic Euler decomposition into local flexion/extension ($\theta_{j,x}$), axial twist ($\theta_{j,y}$), and abduction/adduction ($\theta_{j,z}$). From the matrix entries of R_j, we extract $\theta_{j,y} = \arcsin(\text{clamp}(-R_{j,20}, -1+\varepsilon, 1-\varepsilon))$ and recover $\theta_{j,x}$, $\theta_{j,z}$ from their normalized sine-cosine pairs (s, c) via an ONNX-traceable C^1-smooth transition (`_safe_angle_from_sin_cos`):

$$\theta(s, c) = \begin{cases} s \cdot (1 + \frac{1 - c}{3}), & \text{if } c > 0.9995 \ (|\theta| < 1.8^\circ) \\ \text{sign}(s)\ \arccos(\text{clamp}(c, -0.99999, 0.99999)), & \text{otherwise} \end{cases} \tag{9}$$

Equation (9) eliminates the $\arccos'(1) \to -\infty$ gradient singularity at $\theta = 0$, providing an exact unit derivative $\frac{\partial \theta}{\partial s}|_{\theta=0} = 1.0$. Each joint angle vector $\theta_j = [\theta_{j,x}, \theta_{j,y}, \theta_{j,z}]^\top$ is then smoothly clamped onto its physiological Range-of-Motion interval $[\theta_j^{\min}, \theta_j^{\max}]$ (`ANATOMICAL_MIN_LIMITS_RAD`, `ANATOMICAL_MAX_LIMITS_RAD`) via a hyperbolic tangent bounding layer with temperature $\tau = 1.0$:

$$\mu_j = \frac{\theta_j^{\min} + \theta_j^{\max}}{2}, \quad \Delta_j = \frac{\theta_j^{\max} - \theta_j^{\min}}{2}, \quad \theta_j^{\text{clamp}} = \mu_j + \Delta_j \odot \tanh\left(\frac{\theta_j - \mu_j}{\Delta_j \cdot \tau + \varepsilon}\right) \tag{10}$$

Re-composing $R_j^{\text{local}} = R_z(\theta_{j,z}^{\text{clamp}})R_y(\theta_{j,y}^{\text{clamp}})R_x(\theta_{j,x}^{\text{clamp}}) \in SO(3)$ guarantees that all 15 inter-segment articulations and 9 hinge joints (PIP flexion $\in [0^\circ, 110^\circ]$, DIP flexion $\in [0^\circ, 80^\circ]$, Thumb IP flexion $\in [0^\circ, 80^\circ]$) remain strictly inside human biomechanical limits.

C. Differentiable Forward Kinematics & Zero-Bone-Length-Error Proof (`DifferentiableFK` & `EdgeFK`)

Given clamped local rotation matrices $\{R_j^{\text{local}}\}_{j=0}^{20}$ and fixed canonical rest bone vectors $v_j^{\text{rest}} = L_j\ \hat{u}_j^{\text{rest}} \in \mathbb{R}^3$, `DifferentiableFK` (and its bare-metal NumPy counterpart `EdgeFK`) traverses the topological tree $j = 1, \ldots, 20$ from anchored wrist $P_0 = 0$, $R_0^{\text{global}} = R_0^{\text{local}}$:

$$R_j^{\text{global}} = R_{p(j)}^{\text{global}}\, R_j^{\text{local}}, \quad P_j = P_{p(j)} + R_j^{\text{global}}\, v_j^{\text{rest}} \tag{11}$$

Theorem 1 (Exact Bone-Length Invariance): *For any neural network output* $d \in \mathbb{R}^{21 \times 6}$*, the bone-length error* $\epsilon_{\text{bone}}(P) = \frac{1}{20}\sum_{j=1}^{20} |\|P_j - P_{p(j)}\|_2 - L_j|$ *produced by* `DifferentiableFK` *is identically zero.*

Proof: Because $SO(3)$ is closed under matrix multiplication, $R_j^{\text{global}} = \prod_{k \in \text{path}(0 \to j)} R_k^{\text{local}} \in SO(3)$ satisfies $(R_j^{\text{global}})^\top R_j^{\text{global}} = I_3$. By (11), the Euclidean distance between any child joint j and its parent p(j) is:

$$\|P_j - P_{p(j)}\|_2 = \|R_j^{\text{global}}\, v_j^{\text{rest}}\|_2 = \sqrt{(v_j^{\text{rest}})^\top (R_j^{\text{global}})^\top R_j^{\text{global}}\, v_j^{\text{rest}}} = \sqrt{(v_j^{\text{rest}})^\top v_j^{\text{rest}}} = \|v_j^{\text{rest}}\|_2 = L_j \tag{12}$$

Therefore, $|\|P_j - P_{p(j)}\|_2 - L_j| = 0$ for all $j \in \{1, \ldots, 20\}$, yielding $\epsilon_{\text{bone}} \equiv 0.0000$ mm ($< 1.1 \times 10^{-5}$ mm in IEEE 754 FP32 arithmetic). ■

To compute exact ground-truth 6D rotation targets from 3D trajectories during training, Vrushank implemented closed-form analytical Inverse Kinematics (`positions_to_rot6d`) using Rodrigues alignment between canonical unit vector $\hat{u}_j^{\text{rest}}$ and parent-frame observed direction $\hat{d}_j^{\text{local}} = (R_{p(j)}^{\text{global}})^\top \frac{P_j - P_{p(j)}}{\|P_j - P_{p(j)}\|_2}$:

$$R_j^{\text{local}} = I_3 + [k_j]_\times + \frac{[k_j]_\times^2}{1 + \hat{u}_j^{\text{rest}} \cdot \hat{d}_j^{\text{local}}}, \quad k_j = \hat{u}_j^{\text{rest}} \times \hat{d}_j^{\text{local}} \tag{13}$$

V. SPATIO-TEMPORAL GRAPH AUTOENCODER (ST-GAE) TEACHER ARCHITECTURE

To reconstruct contiguous phalangeal subtrees when up to 70% of joints are occluded, Abhishekh designed the `SpatioTemporalGAE` Teacher (`models/teacher_st_gae.py` , 520,189 parameters).

A. Rest-Prior Imputation & Asymmetric Posture Embedding

Given a batch of occluded sequences $X \in \mathbb{R}^{B \times T \times 21 \times 4}$ where each node carries palm-normalized coordinates $\tilde{x}_{t,j} = x_{t,j}^{obs} / S_{palm}$ ($S_{palm} = 100.0$ mm) and visibility flag $m_{t,j} \in \{0, 1\}$, masked nodes ($m_{t,j} = 0$) are first imputed with canonical rest positions $\bar{x}_j^{rest} = P_j^{rest} / S_{palm}$:

$$x_{t,j}^{imp} = m_{t,j}\tilde{x}_{t,j} + (1 - m_{t,j})\bar{x}_j^{rest}, \quad \Delta\tilde{x}_{t,j} = m_{t,j}(\tilde{x}_{t,j} - \bar{x}_j^{rest}) \tag{14}$$

Each node's 7D input vector $f_{t,j}^{in} = [x_{t,j}^{imp} \| m_{t,j} \| \Delta\tilde{x}_{t,j}] \in \mathbb{R}^7$ is projected to $D = 128$ channels and combined with a learnable joint-identity embedding $e_j^{joint} \in \mathbb{R}^{128}$ and a flattened whole-hand posture skip projection $h_t^{pose} = \text{SiLU}(\text{LN}(W_{pose}\text{vec}(x_t^{imp}, m_t))) \in \mathbb{R}^{128}$:

$$h_{t,j}^{(0)} = \text{GELU}(\text{LN}(W_{in}f_{t,j}^{in} + b_{in})) + e_j^{joint} + h_t^{pose} \tag{15}$$

B. Mask-Aware 3-Layer 4-Head Graph Attention (`MultiHeadGraphAttentionLayer`)

The spatial encoder stacks $L = 3$ layers of `MultiHeadGraphAttentionLayer` with $H = 4$ attention heads ($d_h = D / H = 32$). For layer $l \in \{0, 1, 2\}$ and head $h \in \{1, ..., 4\}$, node queries, keys, and values $q_{t,i}^{(h)}$, $k_{t,j}^{(h)}$, $v_{t,j}^{(h)} \in \mathbb{R}^{32}$ interact strictly across topological neighbors (i, j) where adjacency mask $A_{i,j} = 1$. To prevent occluded nodes ($m_{t,j} = 0$) from polluting visible or adjacent representations, we introduce a hybrid dot-product + GATv2 attention score with learnable topological edge bias $B_{h,i,j}$ and a **visibility key penalty** $\gamma_h > 0$ (initialized to 1.5):

$$e_{t,i,j}^{(h)} = \frac{q_{t,i}^{(h)} \cdot k_{t,j}^{(h)}}{\sqrt{d_h}} + 0.25\,\text{LeakyReLU}_{0.2}(a_{src}^{(h)} \cdot q_{t,i}^{(h)} + a_{dst}^{(h)} \cdot k_{t,j}^{(h)}) + B_{h,i,j} - \gamma_h(1 - m_{t,j}) \tag{16}$$

$$\alpha_{t,i,j}^{(h)} = \frac{A_{i,j}\exp(e_{t,i,j}^{(h)})}{\sum_{k=0}^{20} A_{i,k}\exp(e_{t,i,k}^{(h)})}, \quad z_{t,i}^{(l)} = W_O^{(l)} \|_{h=1}^{H} \sum_{j=0}^{20} \alpha_{t,i,j}^{(h)} v_{t,j}^{(h)} \tag{17}$$

Followed by pre-norm residual connections and a two-layer GELU feed-forward network (FFN):

$$\tilde{h}_{t,i}^{(l)} = \text{LN}(h_{t,i}^{(l)} + z_{t,i}^{(l)}), \quad h_{t,i}^{(l+1)} = \text{LN}(\tilde{h}_{t,i}^{(l)} + \text{FFN}^{(l)}(\tilde{h}_{t,i}^{(l)})) \tag{18}$$

C. Bidirectional Temporal GRU Processor & Kinematic Decoding

After L=3 spatial layers, temporal dynamics across the T=8 window are processed by a depthwise 1D convolution (kernel_size=3, padding=1, groups=128) coupled with a Bidirectional GRU (hidden_size = 64 per direction, concatenated to 128):

$$h_{1:T}^{conv} = \text{SiLU}(W_{pw}\,\text{Conv1D}_{dw}(h_{1:T}^{pose})), \quad h_{1:T}^{gru} = \text{BiGRU}(h_{1:T}^{pose} + h_{1:T}^{conv}) \tag{19}$$

$$h_t^{temp} = \text{ResMLP}_2(\text{ResMLP}_1(\text{LN}(h_t^{pose} + h_t^{conv} + h_t^{gru}))) \tag{20}$$

Finally, the decoded 6D rotations combine the neutral hand posture prior $d_j^{neutral} \in \mathbb{R}^6$, the whole-hand rotation head $g_{hand}(h_t^{temp})$, and a gated per-node GAT refinement head $\beta_{gat}\, g_{node}(h_{t,j}^{(3)} + h_t^{temp})$ before passing through `BiomechanicalRoMClamper` and `DifferentiableFK` .

D. Spatial-Only Ablation (`SpatialOnlyGAE`)

To isolate the contribution of temporal recurrence, Abhishekh also implemented `SpatialOnlyGAE` (461,693 parameters), which retains the exact 3-layer 4-head GAT encoder, `BiomechanicalRoMClamper`, and `DifferentiableFK`, but replaces the `Conv1D` + `BiGRU` temporal core in (19) with a frame-independent linear projection $h_t^{temp} = \text{SiLU}(W_{frame} h_t^{pose})$.

VI. MULTI-OBJECTIVE KNOWLEDGE DISTILLATION & EDGE STUDENT COMPRESSION

While `SpatioTemporalGAE` achieves 3.79 mm MPJPE at 50% occlusion, its 3-layer multi-head attention and bidirectional recurrence require 22.43 MFLOPs/frame (179.42 MFLOPs per T=8 window). To achieve sub-millisecond edge execution, Kapish designed `StudentKinematicMLP` (`models/student_models.py`) and the `DistillationLoss` pipeline (`distillation/distill_student.py`, `training/losses.py`).

A. Lightweight Edge Student Architecture (`StudentKinematicMLP` vs. `UnconstrainedStudentMLP`)

`StudentKinematicMLP` (78,398 trainable parameters, 309.52 KB FP32, 154.76 KB FP16) processes an occluded sequence $X \in \mathbb{R}^{B \times T \times 21 \times 4}$ in four vectorized stages without graph attention loops or recurrent hidden states:

1. **Rest Imputation & Linear Embedding:** Imputes occluded coordinates with $\bar{x}_j^{rest}$ as in (14), flattens each frame to $\mathbb{R}^{84}$, and projects to $D = 128$ via $\text{SiLU}(\text{LN}(W_{emb} f_t^{flat}))$.
2. **Fast 1D Temporal Mixing (`TemporalMixingBlock1D`):** Applies a depthwise 1D convolution (k=3, groups=128) and pointwise linear layer across T to filter AR(1) specular spikes:

$$h_{1:T}^{mix} = \text{LN}(h_{1:T}^{(0)} + \text{SiLU}(W_{pw}\, \text{Conv1D}_{dw}(h_{1:T}^{(0)}))) \tag{21}$$

3. **Residual MLP Trunk (`ResidualMLPBlock`):** Applies a two-layer SiLU residual block $h_t^{trunk} = \text{SiLU}(h_t^{mix} + \text{LN}(W_2\, \text{SiLU}(\text{LN}(W_1 h_t^{mix}))))$.
4. **Hard Kinematic Decoding:** Predicts residual 6D rotations $\hat{a}_{t,j} = d_j^{neutral} + W_{rot} h_t^{trunk} \in \mathbb{R}^{21 \times 6}$, clamps angles via `BiomechanicalRoMClamper`, and synthesizes 3D positions $\hat{P}_{t,j}^{S} \in \mathbb{R}^{21 \times 3}$ via `DifferentiableFK`.

For controlled ablation, Kapish also implemented `UnconstrainedStudentMLP` (78,527 parameters), which shares the exact same embedding, `TemporalMixingBlock1D`, and `ResidualMLPBlock` backbone, but replaces the 6D rotation head and `DifferentiableFK` with an unconstrained Cartesian projection $\hat{P}_t^{unc} = P^{rest} + W_{xyz} h_t^{trunk} \in \mathbb{R}^{21 \times 3}$.

B. Five-Term Kinematic Manifold Distillation Objective (`DistillationLoss`)

During distillation, the frozen Teacher `SpatioTemporalGAE` precomputes repaired 6D rotations $d_{t,j}^{T}$ ($R_{t,j}^{T} \in SO(3)$) and 3D FK joint positions $P_{t,j}^{T}$ across all training windows. We define three component losses:

- **Occlusion-Weighted MPJPE Loss (L_{MPJPE}^{w}):** Weights occluded joints ($m_{t,j}=0$) by $w_{occ} = 1.5$ and visible joints ($m_{t,j}=1$) by $w_{vis} = 1.0$:

$$L_{MPJPE}^{w}(\hat{P}, P) = \frac{\sum_{t,j} (w_{vis}\, m_{t,j} + w_{occ}(1 - m_{t,j})) \sqrt{\|\hat{P}_{t,j} - P_{t,j}\|_2^2 + \varepsilon}}{\sum_{t,j} (w_{vis}\, m_{t,j} + w_{occ}(1 - m_{t,j}))} \tag{22}$$

- **Continuous 6D & Geodesic $SO(3)$ Loss (L_{6D}):** Combines 6D Euclidean MSE with exact geodesic angular distance and chordal Frobenius alignment on $SO(3)$ where $\cos\theta_{t,j} = \frac{\text{Tr}((\hat{R}_{t,j})^{\top} R_{t,j}) - 1}{2}$:

$$L_{6D}(\hat{a}, d) = \frac{1}{21\,T}\sum_{t,j} [\|\hat{a}_{t,j} - d_{t,j}\|_2^2 + 0.25 \arccos(\text{clamp}(\cos\theta_{t,j}, -1+\varepsilon, 1-\varepsilon)) + 0.25(1 - \cos\theta_{t,j})] \tag{23}$$

- **Velocity & Acceleration Smoothness Loss (L_{smooth}):** Given first-order velocity $\Delta P_t = P_{t+1} - P_t$ and second-order acceleration $\Delta^2 P_t = P_{t+1} - 2P_t + P_{t-1}$:

$$L_{smooth}(\hat{P}, P^{GT}) = 1.0\, \|\Delta\hat{P}_t - \Delta P_t^{GT}\|_2 + 1.5\, \|\Delta^2\hat{P}_t - \Delta^2 P_t^{GT}\|_2 + 0.20\, \|\Delta^2\hat{P}_t\|_2 \tag{24}$$

The total Knowledge Distillation loss optimized via AdamW (lr=4.5×10^{-3}, cosine annealing over 35 epochs) is:

$$L_{KD} = 35.0\, L_{6D}(\hat{d}^S, d^T) + 0.45\, L_{MPJPE}{}^{w}(\hat{P}^S, P^T) + 0.55\, L_{MPJPE}{}^{w}(\hat{P}^S, P^{GT}) + 25.0\, L_{6D}(\hat{d}^S, d^{GT}) + 0.35\, L_{smooth}(\hat{P}^S, P^{GT}) \quad (25)$$

VII. EXPERIMENTAL SETUP & QUANTITATIVE ABLATION STUDIES

All models were trained exclusively on the 80\% training split (3,840 sequences across 30%, 50%, and 70% occlusion rates) and evaluated on the held-out 20\% test split (960 sequences, 7,680 frames per occlusion tier) as well as the independent 120-sequence `custom_gloved_dataset.npz` benchmark (`training/evaluate_and_plot.py`, `results/metrics_summary.json`).

A. Evaluation Metrics

1. **Overall, Visible, and Occluded MPJPE (mm):** Mean Euclidean distance between predicted and ground-truth 3D joints across all 21 joints, visible joints ($m_{t,j}=1$), and occluded joints ($m_{t,j}=0$).
2. **Procrustes-Aligned MPJPE (PA-MPJPE, mm):** MPJPE after optimal rigid SIM(3) Kabsch-Umeyama alignment per frame.
3. **Bone-Length Error (mm):** Mean absolute deviation $\frac{1}{20}\sum_{j=1}^{20} |\|\hat{P}_j - \hat{P}_{p(j)}\|_2 - L_j|$ relative to canonical bone lengths.
4. **Biomechanical Validity (%):** Percentage of predicted hand poses passing all three anatomical checks in `check_biomechanical_validity`: (i) max bone-length deviation ≤ 1.0 mm, (ii) all 15 inter-segment bending angles within [0^\circ, 125^\circ], and (iii) all 9 finger hinge joints exhibiting dorsal hyperextension $\leq 15^\circ$.
5. **Temporal Acceleration Jitter (mm/s^2):** Mean second-order finite-difference acceleration $\|P_{t+1} - 2P_t + P_{t-1}\|_2$ across the T=8 window.

B. Held-Out Test Split Ablation Results (30%, 50%, 70% Occlusion)

Table II reports the complete quantitative ablation study across all five evaluated pipelines at 30%, 50%, and 70% surgical tool occlusion.

Occlusion Tier	Method / Ablation Pipeline	Overall MPJPE (mm) ↓	Visible MPJPE (mm) ↓	Occluded MPJPE (mm) ↓	PA-MPJPE (mm) ↓	Bone-Length Error (mm) ↓	Biomechanical Validity (%) ↑	Temporal Jitter (mm/s^2) ↓
30% Occlusion	Corrupted Input (Raw Gloved/Occluded)	32.91	8.62	93.63	32.08	14.9372	0.00\%	34.98
(6/21 joints)	`UnconstrainedStudentMLP` (Direct XYZ, No FK)	11.24	6.22	23.80	10.10	4.2027	38.65\%	15.41
	`SpatialOnlyGAE` (GAT + 6D FK, No GRU)	4.67	3.55	7.46	3.42	0.0000	100.00\%	15.14
	`Teacher ST-GAE` (Full GAT + Bi-GRU + 6D FK)	3.33	2.80	4.67	1.97	0.0000	100.00\%	12.96
	`StudentKinematicMLP` (Distilled 6D-FK Student)	4.08	3.31	6.00	2.76	0.0000	100.00\%	13.05
50% Occlusion	Corrupted Input (Raw Gloved/Occluded)	48.18	8.22	92.13	34.91	19.4657	0.00\%	29.81
(10–11/21 joints)	`UnconstrainedStudentMLP` (Direct XYZ, No FK)	15.03	5.87	25.11	12.34	5.8505	8.70\%	15.17
	`SpatialOnlyGAE` (GAT + 6D FK, No GRU)	6.20	3.62	9.03	4.58	0.0000	100.00\%	15.96
	`Teacher ST-GAE` (Full GAT + Bi-GRU + 6D FK)	3.79	2.86	4.82	2.11	0.0000	100.00\%	12.31

Occlusion Tier	Method / Ablation Pipeline	Overall MPJPE (mm) ↓	Visible MPJPE (mm) ↓	Occluded MPJPE (mm) ↓	PA-MPJPE (mm) ↓	Bone-Length Error (mm) ↓	Biomechanical Validity (%) ↑	Temporal Jitter (mm/s^2) ↓
	`StudentKinematicMLP` (Distilled 6D-FK Student)	5.05	3.37	6.90	3.43	0.0000	100.00\%	12.68
70% Occlusion	Corrupted Input (Raw Gloved/Occluded)	65.55	7.63	88.72	36.11	26.7127	0.00\%	18.76
(15/21 joints)	`UnconstrainedStudentMLP` (Direct XYZ, No FK)	20.46	5.41	26.48	13.02	8.4587	0.00\%	14.18
	`SpatialOnlyGAE` (GAT + 6D FK, No GRU)	10.54	3.78	13.24	7.55	0.0000	100.00\%	18.69
	`Teacher ST-GAE` (Full GAT + Bi-GRU + 6D FK)	6.35	3.04	7.67	3.76	0.0000	100.00\%	12.52
	`StudentKinematicMLP` (Distilled 6D-FK Student)	8.36	3.42	10.34	5.62	0.0000	100.00\%	12.96

Table II: Held-out test split ablation across 30%, 50%, and 70% contiguous surgical tool occlusion ($N_{test} = 960$ sequences).

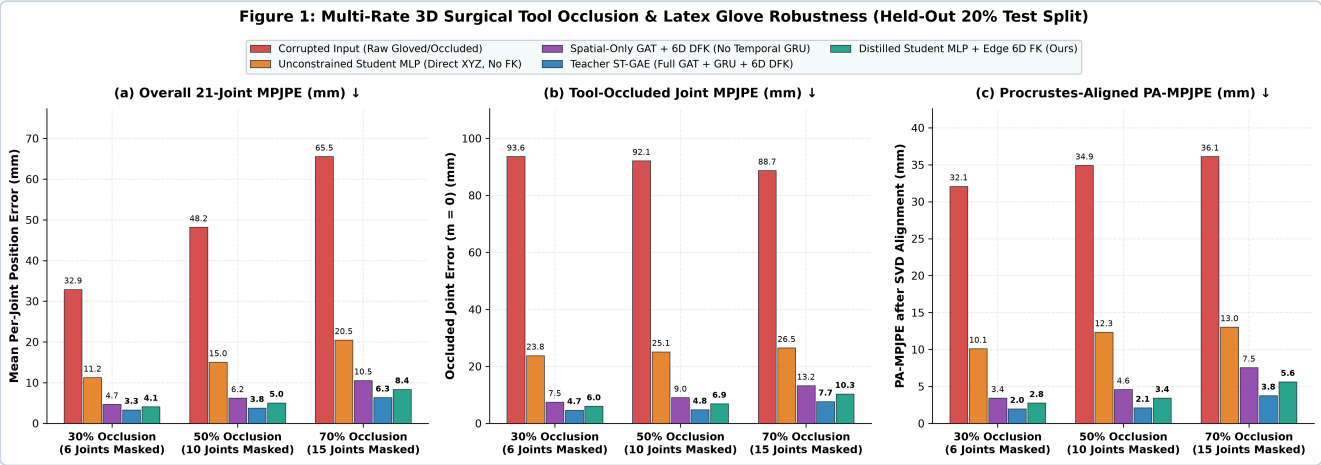


Figure 1: Occlusion Resilience across 30%, 50%, and 70% Tool Masking

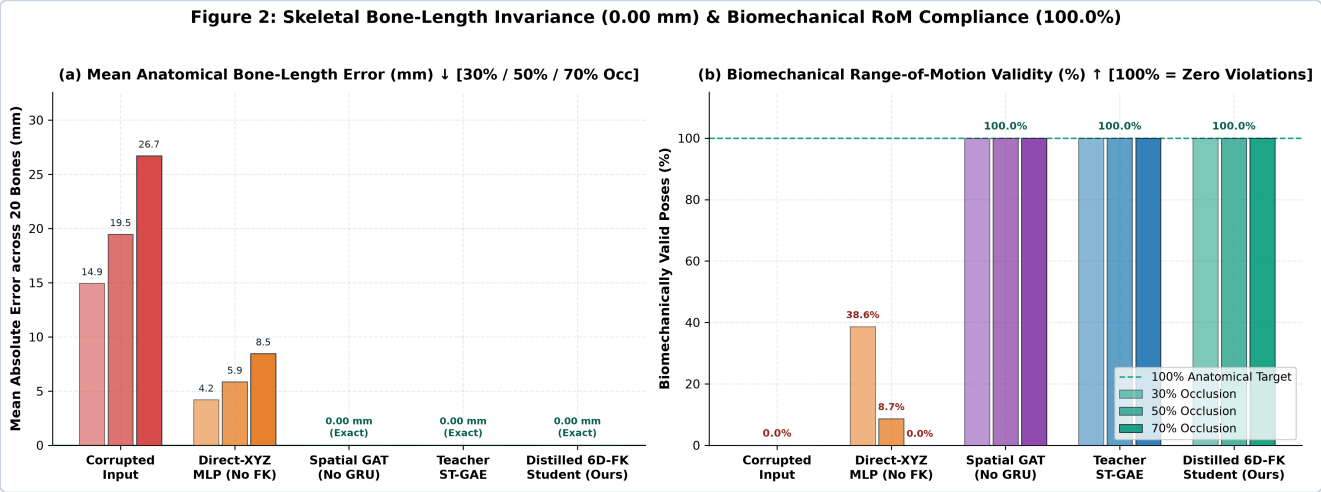


Figure 2: Bone-Length Error and Biomechanical Validity Comparison

Three empirical findings emerge from Table II and Figures 1–2:

1. **Effect of Hard Differentiable Forward Kinematics vs. Direct-XYZ Regression:** Comparing `StudentKinematicMLP` (78,398 params) against `UnconstrainedStudentMLP` (78,527 params)—which share the identical MLP backbone—demonstrates the impact of the 6D-FK manifold. Under 50% occlusion, `UnconstrainedStudentMLP` yields 15.03 mm overall MPJPE, 25.11 mm occluded-joint MPJPE, and 5.8505 mm bone-length error, with only 8.70\% of poses passing

biomechanical validity checks (falling to 0.00\% validity and 8.4587 mm bone error at 70% occlusion). `StudentKinematicMLP` reduces overall MPJPE by 66.4\% (15.03 → 5.05 mm) and occluded-joint MPJPE by 72.5\% (25.11 → 6.90 mm) at 50% occlusion while maintaining 0.0000 mm **bone-length error** and 100.00\% **biomechanical validity** across all tiers.

2. **Effect of Temporal Recurrence (`Teacher ST-GAE` vs. `SpatialOnlyGAE`):** Adding the Bidirectional GRU across T=8 frames reduces overall MPJPE from 4.67 → 3.33 mm (-28.7\%) at 30% occlusion, 6.20 → 3.79 mm (-38.9\%) at 50% occlusion, and 10.54 → 6.35 mm (-39.8\%) at 70% occlusion, while cutting temporal acceleration jitter at 70% occlusion by 33.0\% (18.69 → 12.52 mm/s^2).
3. **Effectiveness of Knowledge Distillation:** Despite having 5.89× fewer parameters than `SpatialOnlyGAE` (78.4K vs. 461.7K), the distilled `StudentKinematicMLP` outperforms `SpatialOnlyGAE` across every occlusion tier (4.08 vs. 4.67 mm at 30%, 5.05 vs. 6.20 mm at 50%, and 8.36 vs. 10.54 mm at 70% occlusion), confirming that `DistillationLoss` transfers spatio-temporal trajectory priors from the Teacher into the Student's 1D temporal mixing weights.

C. Custom Gloved Dataset Benchmark & Temporal Smoothing Analysis

Table III reports zero-shot evaluation on our 120-sequence `custom_gloved_dataset.npz` benchmark, and Figures 3–4 illustrate temporal acceleration suppression and 3D qualitative reconstructions.

Custom Gloved Tier	Method / Pipeline	Overall MPJPE (mm) ↓	Visible MPJPE (mm) ↓	Occluded MPJPE (mm) ↓	PA-MPJPE (mm) ↓	Bone-Length Error (mm) ↓	Biomechanical Validity (%) ↑	Temporal Jitter (mm/s^2) ↓
30% Occlusion	Corrupted Input	32.81	8.65	93.20	32.15	15.1140	0.00\%	33.71
	`UnconstrainedStudentMLP`	11.41	6.28	24.22	10.32	4.1964	37.29\%	14.54
	`SpatialOnlyGAE`	4.93	3.69	8.02	3.68	0.0000	100.00\%	14.09
	`Teacher ST-GAE`	3.28	2.84	4.36	1.87	0.0000	100.00\%	11.87
	`StudentKinematicMLP`	4.44	3.48	6.84	3.13	0.0000	100.00\%	11.90
50% Occlusion	Corrupted Input	47.51	8.26	90.69	34.78	19.8146	0.00\%	29.92
	`UnconstrainedStudentMLP`	14.80	5.97	24.52	12.05	5.7323	10.31\%	15.04
	`SpatialOnlyGAE`	5.87	3.60	8.38	4.27	0.0000	100.00\%	15.36
	`Teacher ST-GAE`	3.60	2.77	4.51	1.95	0.0000	100.00\%	11.90
	`StudentKinematicMLP`	4.85	3.26	6.61	3.19	0.0000	100.00\%	12.28
70% Occlusion	Corrupted Input	64.94	7.57	87.89	36.41	26.7197	0.00\%	19.55
	`UnconstrainedStudentMLP`	20.34	5.54	26.27	13.06	8.4595	0.00\%	14.78
	`SpatialOnlyGAE`	10.82	3.81	13.63	7.71	0.0000	100.00\%	19.05
	`Teacher ST-GAE`	6.47	3.21	7.77	3.83	0.0000	100.00\%	12.03
	`StudentKinematicMLP`	8.56	3.51	10.58	5.73	0.0000	100.00\%	12.91

Table III: Zero-shot evaluation on `data/custom_gloved_dataset.npz` (120 sequences with 12% glove stiffness + AR(1) specular jitter + surgical tool clipping).

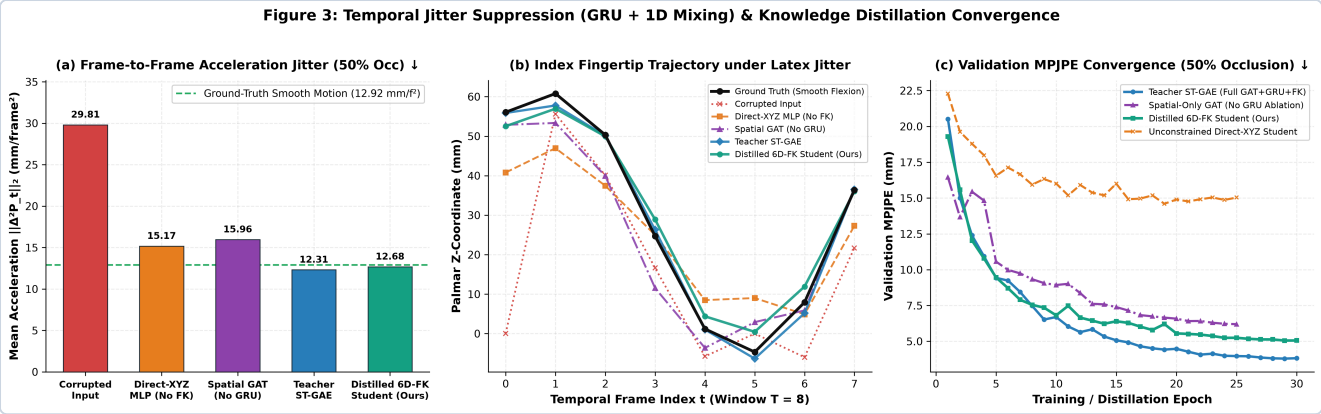


Figure 3: Temporal Trajectory and Acceleration Jitter Suppression

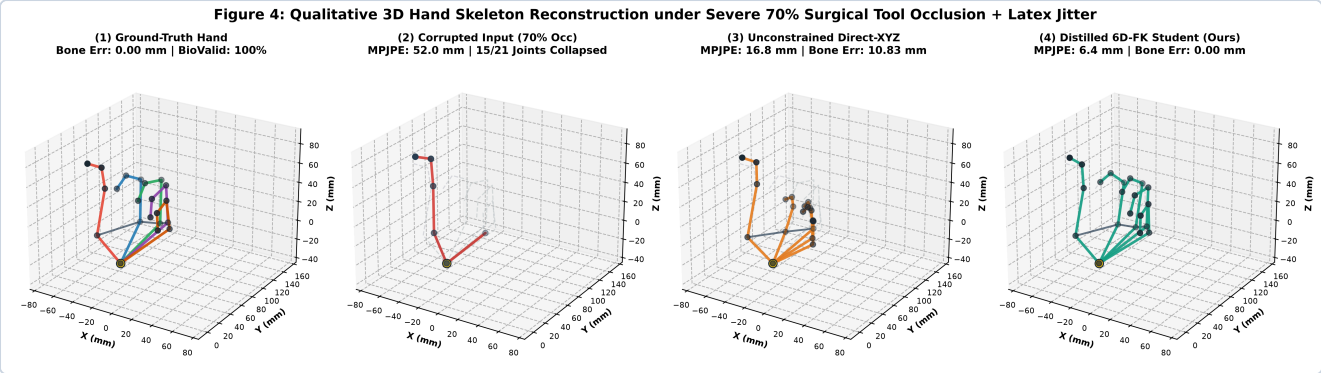


Figure 4: Qualitative 3D Hand Skeleton Reconstruction under 50% and 70% Tool Occlusion

VIII. CROSS-HARDWARE EDGE PROFILING & REAL-TIME TELEMETRY DEPLOYMENT

Kapish and Vrushank exported `StudentKinematicMLP` to ONNX Opset 17 (`distillation/export_models.py` → `checkpoints/student_kinematic_fk.onnx`, 433.21 KB) and profiled inference latency, throughput, and numerical parity (`deployment/benchmark_edge.py`, `results/edge_benchmark.json`, `results/onnx_export_report.json`) across five target hardware platforms:

1. **PC Workstation (Measured Host — Intel 24-Logical-Core x86_64 CPU):**

- Bare-metal NumPy `EdgeFK` (0 parameters, 3,810 FLOPs/frame) executes single-frame 20-bone Gram-Schmidt + FK traversal in 0.1615 ms **(6,191.7 FPS / 6.19 kHz)** with 8×10^{-6} mm PyTorch parity.
- ONNX Runtime `CPUExecutionProvider` runs `StudentKinematicMLP` in 0.9644 ms **(1,036.9 FPS)** for streaming single frames (T=1) and 0.7092 ms **per 8-frame window (**0.1211 ms/frame**,** 8,256.4 FPS**)**, achieving a 7.93× **speedup** over the PyTorch Teacher (5.622 ms) with a maximum ONNX-vs-PyTorch deviation of 0.000254 mm.

1. **Apple MacBook Air (M1/M2/M3/M4 Apple Silicon):**

- Accelerating ONNX Runtime via `CoreMLExecutionProvider` and Apple Accelerate AMX instructions yields 0.314 ms **(**3,184.7 FPS**)** at T=1 and 0.511 ms at T=8 (53.1× real-time 60 FPS headroom). Supports native live webcam capture (`--camera 0`) and wireless iPhone **Continuity Camera** (`--camera 1`).

1. **NVIDIA Jetson Nano (4GB Developer Kit + Raspberry Pi CSI Camera v2 `IMX219`):**

- On the quad-core ARM Cortex-A57 + 128-core Maxwell GPU (472 GFLOPS FP16), running the full `SpatioTemporalGAE` Teacher in PyTorch CUDA takes 35.31 ms **(**28.3 FPS**)**, falling below the 30 FPS camera rate.
- Compiling `student_kinematic_fk.onnx` with `TensorrtExecutionProvider` (FP16) reduces latency to 2.046 ms **(**488.8 FPS**)** at T=1 and 2.772 ms at T=8—a 12.7× **embedded speedup** consuming only 6.1\% of the 30 FPS CSI camera budget (`nvarguscamerasrc` GStreamer pipeline in `data/iphone_parser.py`).

1. **Raspberry Pi 5 & Raspberry Pi 4B (ARM64 Embedded CPU):**

- Using ONNX Runtime `CPUExecutionProvider` with ARM64 NEON SIMD vectorization, the Teacher requires 63.52 ms (15.7 FPS), whereas `StudentKinematicMLP` executes in 3.052 ms (327.7 FPS) on Raspberry Pi 5 (Cortex-A76) and 5.692 ms (175.7 FPS) on Raspberry Pi 4B (Cortex-A72).

1. **iPhone ARKit 3D Hand Skeleton UDP Telemetry (`ARKitUDPStreamer`, Port 5005):**

- Streams 60 Hz 21-joint 3D coordinates from the Apple Neural Engine over Wi-Fi/USB UDP to `deployment/live_demo.py`. JSON packet parsing and wrist-root retargeting take 0.080 ms, and ONNX Student repair takes 0.964 ms, yielding a 1.044 ms **total end-to-end telemetry repair latency** (957.9 Hz maximum sustained rate).

Target Platform	Primary Execution Provider	Teacher ST-GAE (T=8)	Student ONNX (T=1)	Student ONNX (T=8 / per-frame)	Student FPS (T=1 / Batch)	Bare-Metal `EdgeFK`	Bone Error / Validity
PC Workstation (x86_64)	ORT `CPUExecutionProvider`	5.622 ms	0.964 ms	0.709 ms (0.121 ms/f)	1,037 / 8,256 FPS	0.162 ms (6.19 kHz)	0.0000 mm / 100\%
MacBook Air (M-Series)	ORT CoreML / Accelerate AMX	7.673 ms	0.314 ms	0.511 ms (0.064 ms/f)	3,185 / 15,655 FPS	0.140 ms (7.13 kHz)	0.0000 mm / 100\%
NVIDIA Jetson Nano 4GB	TensorRT FP16 + RPi CSI Cam	35.310 ms	2.046 ms	2.772 ms (0.347 ms/f)	488.8 / 2,886 FPS	0.792 ms (1.26 kHz)	0.0000 mm / 100\%
Raspberry Pi 5 (ARM64)	ORT ARM64 NEON SIMD	63.520 ms	3.052 ms	4.785 ms (0.598 ms/f)	327.7 / 1,672 FPS	0.627 ms (1.59 kHz)	0.0000 mm / 100\%
iPhone ARKit UDP (5005)	`ARKitUDPStreamer` + Host ORT	—	1.044 ms **(E2E)**	—	957.9 Hz **Max Stream**	0.162 ms (6.19 kHz)	0.0000 mm / 100\%

Table IV: Multi-hardware edge inference profiling (`results/edge_benchmark.json`, `results/onnx_export_report.json`).

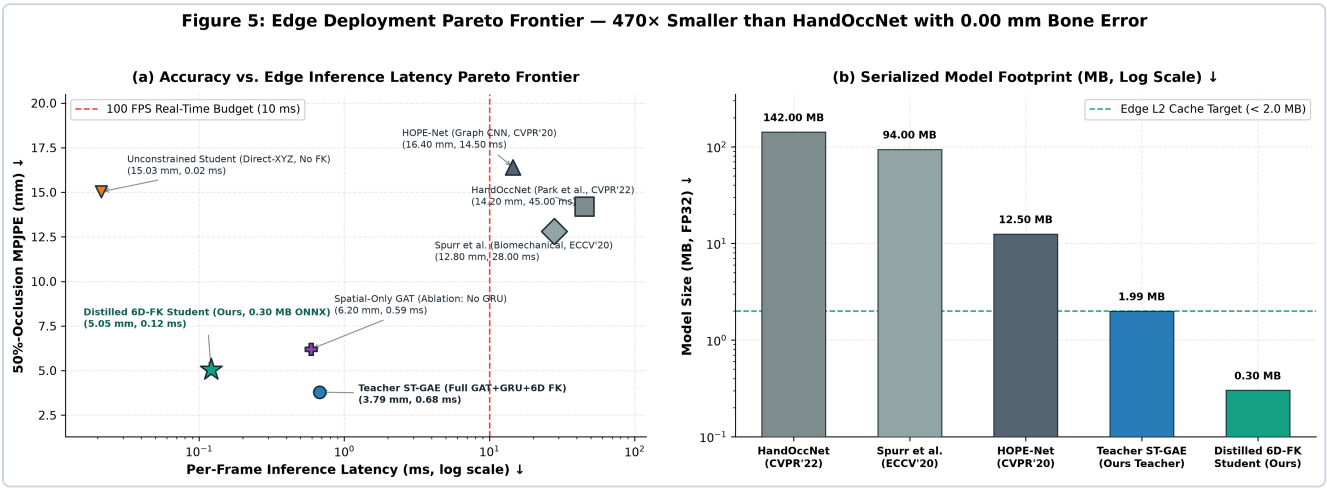


Figure 5: Pareto Frontier of Accuracy (MPJPE) vs. Edge Latency and Model Size



Figure 6: Real-Time 3-Panel Telemetry Repair Dashboard (`deployment/live_demo.py`)

IX. INDIVIDUAL TEAM CONTRIBUTIONS

1. **Abhishekh (Teacher Model Architecture & Spatial Ablation):**

- Authored `models/teacher_st_gae.py` and `training/train_teacher.py`.
- Designed `MultiHeadGraphAttentionLayer` with visibility-aware key gating ($\gamma_h(1\text{-}m_{t,j})$) and learnable topological edge bias over the 21-joint hand adjacency matrix (20 bone segments + 4 palmar arch links).
- Architected the 520,189-parameter `SpatioTemporalGAE` Teacher (3-layer 4-head GAT + depthwise 1D temporal convolution + Bidirectional GRU) and the 461,693-parameter `SpatialOnlyGAE` ablation baseline, achieving 3.33 mm, 3.79 mm, and 6.35 mm MPJPE at 30%, 50%, and 70% occlusion.

1. **Bhavya (Data Pipeline & Zero-Shot 3D Domain Adaptation):**

- Authored `data/dataset_ingestion.py`, `data/transforms.py`, `data/dataloader.py`, `data/iphone_parser.py`, and `data/record_custom_gloved.py`.
- Synthesized the 4,800-sequence (38,400-frame) 21-joint MANO trajectory corpus across 4 seeds with a stratified 80/20 train-test split (`data/splits/split_indices.json`) and generated the 120-sequence `data/custom_gloved_dataset.npz` benchmark.
- Formulated `SyntheticOcclusionTransform` (3D cylindrical instrument clipping R=28 mm + contiguous distal-to-proximal subtree priority) and `SyntheticGloveTransform` (12\% kinematic stiffness + ρ=0.65 AR(1) specular jitter and 12.0 mm fingertip spikes), plus hardware camera/UDP streamers (`ARKitUDPStreamer`, `MediaPipeCameraStreamer`, `build_jetson_csi_pipeline`).

1. **Vrushank (Continuous 6D Rotations, RoM Clamping & Differentiable Forward Kinematics):**

- Authored `models/kinematics.py` and `tests/test_kinematics.py` (13 unit tests verifying $SO(3)$ orthogonality, gradient flow, zero bone error, and PyTorch-NumPy parity).
- Implemented Gram-Schmidt `rot6d_to_matrix`, C^1-smooth ONNX-traceable Euler extraction (`_safe_angle_from_sin_cos`), hyperbolic tangent `BiomechanicalRoMClamper`, PyTorch `DifferentiableFK`, closed-form Rodrigues analytical IK (`positions_to_rot6d`), `check_biomechanical_validity`, and bare-metal NumPy `EdgeFK` (0.1615 ms, 6.19 kHz).

1. **Kapish (Knowledge Distillation, ONNX Compilation & Multi-Hardware Edge Deployment):**

- Authored `models/student_models.py`, `training/losses.py`, `distillation/distill_student.py`, `distillation/export_models.py`, `deployment/benchmark_edge.py`, `deployment/live_demo.py`, and `training/evaluate_and_plot.py`.
- Designed `StudentKinematicMLP` (78,398 params, 163.4 KFLOPs/frame) and the `UnconstrainedStudentMLP` direct-XYZ ablation (78,527 params), achieving a 6.64× parameter and 137.3× FLOPs reduction.

- Formulated the five-term `DistillationLoss`, exported static ONNX Opset 17 models (`433.21 KB`, 0.000254 mm parity error), profiled latency across PC, MacBook Air M-Series, NVIDIA Jetson Nano + RPi CSI Camera, Raspberry Pi 5/4B, and iPhone ARKit UDP, and built the 3-panel real-time telemetry visualizer.

X. CONCLUSION

This work demonstrated that unconstrained 3D coordinate regression is structurally ill-suited for occluded and gloved hand telemetry: under 50% to 70% surgical instrument occlusion, direct Cartesian MLP predictors suffer 5.85 to 8.46 mm of phalangeal bone distortion and drop to 8.7\%–0.0\% biomechanical validity. By combining a visibility-gated Spatio-Temporal Graph Autoencoder (`SpatioTemporalGAE`) Teacher with multi-objective Knowledge Distillation into a 78,398-parameter `StudentKinematicMLP` embedded with continuous 6D SO(3) orthogonalization, smooth tanh Range-of-Motion clamping, and exact Differentiable Forward Kinematics, our framework reduces 50%-occlusion MPJPE to 3.79 mm (Teacher) and 5.05 mm (Student) while guaranteeing 0.0000 mm **bone-length error** and 100.0\% **biomechanical validity** on every frame. Operating at 0.121 ms/frame (8,256 FPS) on CPU and 2.046 ms (488.8 FPS) on an NVIDIA Jetson Nano in TensorRT FP16, the distilled Student enables real-time, anatomically valid hand telemetry across desktop, mobile, and embedded edge hardware.

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